

Global and Transdisciplinary Perspectives on Dark Patterns and Deceptive Design Practice

ダークパターンとひとをだますデザインに関する
国際的かつ学際的な交流

Katie Seaborn
Department of Industrial Engineering
and Economics
Institute of Science Tokyo
Tokyo, Japan
seaborn.k.aa@m.titech.ac.jp

Colin M. Gray
UXP2 Lab
Indiana University
Bloomington, Indiana, USA
comgray@iu.edu

Johanna Gunawan
Law and Tech Lab
Maastricht University
Maastricht, Netherlands
johanna.gunawan@maastrichtuniversity.nl

Thomas Mildner
University of Bremen
Bremen, Germany
mildner@uni-bremen.de

René Schäfer
RWTH Aachen University
Aachen, Germany
rschaefer@cs.rwth-aachen.de

Lorena Sanchez Chamorro
Human-Computer Interaction
Research Group
University of Luxembourg
Esch-Sur-Alzette, Luxembourg
lorena.sanchezchamorro@uni.lu

Satoshi Nakamura
Meiji University
School of Interdisciplinary
Mathematical Sciences
Nakano, Tokyo, Japan
nkmr@meiji.ac.jp

Abstract

Dark patterns and deceptive designs (DPs) refer to user interfaces (UIs) that trick people into interactions that benefit the service providers. Today, academic research, legal action, and media coverage has raised awareness among a diversity of stakeholders worldwide. Yet, the lens has focused on Western and English contexts. We propose a Special Interest Group (SIG) that centres on cross-cultural and interdisciplinary engagement. The organizing team, who hail from a plurality of nations and disciplines, will spark discussion by sharing their knowledge—findings, frameworks, methods, and tools—and culturally-sensitive perspectives on deception in modern digital products and services. Attendees will participate in a small group drawing activity, whereby culturally-specific DPs and disciplinary perspectives can be surfaced and communicated without reliance on a specific language or cultural frame. This SIG is expected to draw in a diversity of designers, researchers, security experts, and legal scholars concerned about ethical design practice.

CCS Concepts

• **Human-centered computing** → *HCI theory, concepts and models*; Collaborative and social computing theory, concepts and paradigms;
• **Interaction design theory, concepts and paradigms**; • **Security and privacy** → *Human and societal aspects of security and privacy*;
• **Social and professional topics** → **Codes of ethics**; *Privacy policies*; Governmental regulations; Computer crime; Government technology policy; *Cultural characteristics*; • **Applied computing** → *Law*.

Keywords

Dark Patterns, Deceptive Design, Manipulative User Interfaces, Ethics, Regulation, ダーク・パターン, 欺瞞的デザイン, 人をだますデザイン

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3716294>

ACM Reference Format:

Katie Seaborn, Colin M. Gray, Johanna Gunawan, Thomas Mildner, René Schäfer, Lorena Sanchez Chamorro, and Satoshi Nakamura. 2025. Global and Transdisciplinary Perspectives on Dark Patterns and Deceptive Design Practice: ダークパターンとひとをだますデザインに関する 国際的かつ学際的な交流. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3716294>

1 Motivation & Background

Dark patterns¹ and deceptive designs (DPs) have become a feature of digital media and commercial software the world over. These user interface (UI) elements trick, coerce, or otherwise force the user to take actions that have hidden consequences, generally to the benefit of shareholders over users [22, 25, 29, 40]. DPs are a *transdisciplinary* challenge, touching on matters of interaction design (IxD) and user experience (UX) on the human-computer interaction (HCI) side, as well as law and governance, both practical and scholarly, and matters of ethics, in design and industrial practice. Many DPs are explained by psychology, with explanatory mechanisms that can be traced back to the social sciences and behavioural economics [8, 60]. A growing body of research and activism led by HCI scholars and others has raised the issue of DPs to the fore. DPs and their ilk have been found in e-commerce websites and apps [26, 30, 32, 39, 47], consent banners [23, 28, 38], video games [1, 13, 63], social media [41, 44, 46, 57], conversational agents and interfaces [42, 52], augmented and virtual reality apps [11, 61], and Internet-of-Things (IoT) offerings [34]. Foundational work has contributed ontologies and taxonomies [22, 25, 48], operationalizations [40], prevalence rates [10, 30, 35, 39], and the impact on users [6, 7, 15, 28, 44, 53]. Academia, industry, and the law are all responding to the ethical issues raised across these diverse areas of work.

Duplicity in design is also a *global* concern. Worldwide interest in DPs has surged in recent years, as represented in the international reports by the world-leading Organisation for Economic Co-operation and Development (OECD) [48, 49]. The Japan-led 2022 OECD report raised the nation-transcendent issue of consumer vulnerability and policy [48]. Awareness and action is also occurring at regional and local scales [5, 9, 30, 41]. In Europe, the European Data Protection Board (EDPB) [12] has led efforts to take action on DPs, starting by crafting guidelines for social media users. In Latin America, work has begun to critically analyze DPs in social media [41]. In Asia, new legislation and guidelines have been adopted or proposed in India [50, 51], Korea [3], and Japan [2]. The prevalence of DPs across nations has become plain. The 2022 EU Commission report found a rate of at least one pattern across 97% of popular apps and websites for EU consumers [37]. Foundational work by Di Geronimo et al. [10] found that 95% of 240 US Google Play Store apps contained at least one DP. Later, Hidaka et al. [30] confirmed similar results for the Japanese store, finding 93.5% in the top 200 apps. These studies demonstrate the global threats posed by DPs for users of most popular apps. Governmental organizations and legislative bodies have begun to take note, such as the warning in 2023 from the Japanese Consumer Affairs division the conflation of the Japanese yen symbol and the Chinese yuan². Yet, most work and collaboration has taken place in industrialized Western and Anglocentric nations [58].

¹We use this term to ensure that our efforts connect with relevant scholarship and legal statutes. We acknowledge that the ACM Diversity and Inclusion Council now includes the term “dark patterns” on a list of problematic terms (<https://www.acm.org/diversity-inclusion/words-matter>). At present, other terms, such as “deceptive design,” “disloyal design,” and “manipulative user interfaces” remain underused in academic and regulatory circles. In Japan, the media’s reliance on a direct translation of the term “dark pattern” as ダーク・パターン (daku patān) has left most unfamiliar with other terms, like deceptive design or 欺瞞的デザイン (giman-teki-dezain). We use the abbreviation “DP” as a compromise.

²https://www.kokusen.go.jp/news/data/n-20230419_2.html (Note: Japanese-only)

In this SIG, we seek to address these gaps and opportunities under the banner of *global and transdisciplinary engagement*. We are inspired by CHI ‘25 being held in Japan, an Eastern nation that represents an emerging global player for academic, design, and regulatory action on DPs. The time is ripe for syncing local and international efforts on ethical design practice and governance in a transdisciplinary and translational manner.

2 Aims of the SIG

Tackling the global challenge of DPs is a culturally-sensitive and transdisciplinary effort. The wealth of HCI scholarship, case studies on lawsuits and legal action, and reports from regulatory bodies the world over signal unquestionable engagement on the matter from all fronts. This body of work has been used to push for governance and legal mandates aiming to protect consumers and guide industry away from interactive designs that undermine user choice and agency. Notably, the CHI community has led and established a nexus for engagement within this dynamic space.

The proposed SIG represents the next phase in a series of successful engagements on the topic of DPs within and beyond CHI. The CHI ‘21 workshop [36] led to the creation of a small international and transdisciplinary community on Slack. A general SIG [24] raised DPs to the forefront of CHI affairs and a complementary panel [17] drew over 100 attendees at CHI ‘23. The CHI ‘24 workshop [19] centred on action at all scales: scholarship, governance, legal, industrial, and more. This workshop revealed crucial new challenges and opportunities related to *cultural plurality* and *transdisciplinarity*. Recognizing that a panel format may be prescriptive and a workshop may only draw those identifying as DP scholars, we are proposing an interactive and inclusive SIG engagement.

In this SIG, we aim to bring in new voices and perspectives on DPs by addressing two core dimensions. First, *geography*: we purposely seek to expand the purview of DP work beyond its Euro- and US-centered core, particularly by developing a broader global community. Second, *discipline*: we invite a range of HCI and adjacent practitioners who have a stake in DPs or can provide novel frames of reference. To this end, the SIG has two core objectives:

- (1) **Knowledge-sharing:** We seek to share knowledge on the diversity of frameworks, methods, and tools used to understand the prevalence and impact of DPs on a variety of stakeholders. We aim to establish a global understanding of key concepts and theories (e.g., values, trust, deception), modes of practice, and perspectives on ethics across cultures and disciplines. We seek to link disparate knowledge bases and find commonalities. We expect topics to include technical detection, user awareness, user impact, expert values, expert practice, legal case studies, and more. We will centre work that brings in culturally-sensitive epistemological frames and ontologies, highlights regional operational differences in design practice, and unearths local varieties of DPs.
- (2) **Taking action:** We aim to generate solutions to DPs in a culturally-sensitive and collaborative way. The SIG will enable disparate parties to meet and combine forces on the

creation of versatile and relevant solutions to DPs. We expect work on design strategies, educational platforms, cross-cultural initiatives, participatory design with consumer, industry, and/or legal representation, and more. We will particularly encourage work that seeks to develop international teams or identifies strategies relevant to global or localized design or regulatory efforts. This year, we have included an interactive activity to help facilitate this objective.

These aims will be achieved through pre- and post-activity **organizer-led discussions**—to set the stage, open up the problem space, and surface themes—alongside an interactive small group **activity centred on drawing methodology** [14], which allows language-free communication and is particularly suited to exploring UI phenomena.

As a starting point, we offer the following launching points, with two caveats: (i) we do not wish to be prescriptive and (ii) we expect novel topics to emerge as a direct result of our concerted global and transdisciplinary focus.

- How are DPs characterized within particular regional, cultural, and/or disciplinary contexts? How do users, experts, and other shareholders perceive DPs? Are there famous examples?
- What terms and words are used to describe DPs in different disciplines and languages?
- What is your personal connection to DPs? What experience/s led to your interest in DPs?
- What frameworks are used to describe deceptive design practices? What terms exist? What language is used when describing DPs? What insights can we glean from translating words among languages?
- What taxonomies are used to describe DPs? Have regional varieties of DPs been discovered?
- What theories are used to understand how DPs work and what impact they can have on users?
- What methods—academic, industrial, designerly, legal, governmental, social—are used to address DPs? Are there, for instance, informal communities banding around hashtags outside of English-speaking social platforms?
- What values and codes of ethics are leveraged in discussions about UI-based digital manipulation?
- What expert and industrial perspectives exist? What consumer and user perspectives exist? What other stakeholders are there? Are there cultural components? Underrepresented disciplinary frames?
- What is HCI research on DPs missing?
- What action is being taken in certain regions? What has been the cultural response to legal and social pushback against DPs? What can we learn from legal case studies in non-Western regions?
- What solutions have been explored? What strategies have been successful? Are these technical, legal, industrial, social? What players and entities have been involved?
- What laws, regulations, legal mandates, and governance has been developed around DPs? How has this affected design and industrial practice (or not)? What new issues have arisen?
- What linkages are missing: locally and internationally? Within and cross-discipline? Where might there be opportunities for cross-pollination and achieving the critical mass needed for effective change at a global scale?

3 Organizers

The organizing team comprises researchers in the local Japanese and global research communities. We have contributed to building a dynamic space that crosses the regulatory, design, and HCI communities. Our track record includes several successful workshops, notably “Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices” at CHI ‘24 [19], EduCHI at CHI ‘23 [21], manipulative social agents at CUI ‘23 [4], and “What Can CHI Do About Dark Patterns?” at CHI ‘23 [36]. We are deeply embedded within the field and actively contribute to DP scholarship. We comprise mid-career scholars, early-career researchers, and doctoral students.

- **Katie Seaborn** is an Associate Professor at Institute of Science Tokyo (formerly Tokyo Institute of Technology) in Japan. In 2022, their team discovered novel patterns tied to the Japanese socio-linguistic context [30]. In 2023, they collaborated with Japanese public broadcaster NHK to run the first user study with Japanese consumers on these novel patterns [59]. As a recognized leader in Japan, Professor Seaborn’s work on DPs has been covered by NHK, Yomiuri Shimbun, and other major outlets for Japanese television and print media.
- **Colin M. Gray** is an Associate Professor at Indiana University (USA), where they serve as program director for a graduate program in HCI/d. Their research has contributed foundational knowledge on DPs [20], including a unified ontology [22, 25]; connections between design, law, and HCI [23]; and articulation of challenges in supporting ethical design practices [16, 18].
- **Johanna Gunawan** is an Assistant Professor of Computer Science and Law at the Maastricht University Law and Tech Lab (Netherlands). Her interdisciplinary work spans security/privacy, HCI, design, and law, focused on understanding dark patterns and power asymmetry in ubiquitous or emergent technologies [26, 34, 52] and policy considerations for improving consumer protections against DPs and similar designs [27, 28].
- **Thomas Mildner** is a postdoctoral researcher at the University of Bremen, Germany. With a specific focus on DPs in social media [44, 46], Thomas’ research concerns (un-)ethical design in digital interfaces [42, 43, 45], methods to analyze DPs, and the development of countermeasures to support autonomous decision-making.
- **René Schäfer** is a PhD candidate at the RWTH Aachen University in Germany. His research focuses on vulnerable groups [56], how companies hinder users in exercising their rights using DPs [35], and visual countermeasures to help people deal with malicious and deceptive designs [54, 55].
- **Lorena Sánchez** is a postdoctoral researcher at the University of Twente (Netherlands). Her research focuses on the intersection of manipulative design and vulnerability: how

to protect users by increasing their resilience. She aims to create multidisciplinary solutions with a social perspective.

- **Satoshi Nakamura** is a Professor at Meiji University in Japan. He is interested in how user selection in choice-based interfaces can be influenced, potentially as DPs. His research focuses on how interface elements, such as pop-out effects, font styles, position, device type, and progress bars, can affect user selections [31, 33, 60, 62].

4 Expected Outcomes & Next Steps

After the SIG, we will focus on creating and sustaining global and translational communication among participating researchers, designers, legal scholars, practitioners, and regulators hailing from their home nations. We will expand membership in the central Slack workspace, share networks to connect community members, and raise awareness of local and global events and engagements on combating DPs.

In the SIG, we will discuss how to encourage communication among the global community of HCI scholars, practitioners in the adjacent sciences and humanities, and members of the legal and regulatory communities. We will gather ideas on how to proactively support DP-free design internationally. We will address emergent governance and law across the East (e.g., in Japan [2], India [50, 51], Korea [3]), extant rules or enforcement in the US and EU, and work in regions with less established guidance or practice. We will use the Slack group to link like-minded activists. We will encourage attendees to be listed on the website as regional points of contact.

The organizers will aim to publish a short article in *Interactions* magazine to disseminate the results of the SIG, as well as highlight opportunities, methods, and suggestions for international and transdisciplinary action. Selected drawings (with permission) will be showcased on the SIG website:

> <https://chi2025.darkpatternsresearchandimpact.com>

These outcomes and opportunities will be disseminated through the same channels used for pre-SIG recruitment, aiming to highlight the growing diversity of the community and bring in new voices for future events.

Acknowledgments

This SIG will be funded in part by a JST PRESTO · さきがけ grant (#JPMJPR24I6).

References

- [1] Jacob Aagaard, Miria Emma Clausen Knudsen, Per Bækgaard, and Kevin Doherty. 2022. A Game of Dark Patterns: Designing Healthy, Highly-Engaging Mobile Games. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. ACM, New York, NY, USA, 1–8. doi:10.1145/3491101.3519837
- [2] Consumer Affairs Agency. n.d.. Consumer Research and International Affairs. https://www.caa.go.jp/en/policy/consumer_research/
- [3] Korea Consumer Agency. 2024. 75.7% of Subscription Service Websites and Mobile Apps Worldwide Use Dark Patterns. <https://www.kca.go.kr/eng/sub.do?menukey=6007&mode=view&no=1003719829>
- [4] Vito Avanesi, Johanna Rockstroh, Thomas Mildner, Nima Zargham, Leon Reicherts, Maximilian A. Friehs, Dimosthenis Kontogiorgos, Nina Wenig, and Rainer Malaka. 2023. From C-3PO to HAL: Opening The Discourse About The Dark Side of Multi-Modal Social Agents. In *Proceedings of the 5th International Conference on Conversational User Interfaces* (Eindhoven, Netherlands) (CUI '23, Article 62). Association for Computing Machinery, New York, NY, USA, 1–7. doi:10.1145/3571884.3597441
- [5] Luiz Adolpho Baroni, Alisson Andrey Puska, Luciana Cardoso de Castro Salgado, and Roberto Pereira. 2021. Dark Patterns: Towards a Socio-Technical Approach. In *Proceedings of the XX Brazilian Symposium on Human Factors in Computing Systems* (Virtual Event, Brazil) (IHC '21). Association for Computing Machinery, New York, NY, USA, Article 15, 7 pages. doi:10.1145/3472301.3484336
- [6] Kerstin Bongard-Blanchy, Arianna Rossi, Salvador Rivas, Sophie Doublet, Vincent Koenig, and Gabriele Lenzini. 2021. "I am Definitely Manipulated, Even When I am Aware of it. It's Ridiculous!" - Dark Patterns from the End-User Perspective. In *Designing Interactive Systems Conference 2021* (Virtual Event, USA) (DIS '21, Vol. 1). Association for Computing Machinery, New York, NY, USA, 763–776. doi:10.1145/3461778.3462086
- [7] Ida Borberg, Rene Hougaard, Willard Rafnsson, and Oksana Kulyk. 2022. "So I Sold My Soul": Effects of Dark Patterns in Cookie Notices on End-User Behavior and Perceptions. In *Usable Security and Privacy (USEC) Symposium 2022*. NDSS, San Diego, CA, USA, 11 pages. doi:10.14722/usec.2022.23026
- [8] Weichen Joe Chang, Katie Seaborn, and Andrew A. Adams. 2024. Theorizing Deception: A Scoping Review of Theory in Research on Dark Patterns and Deceptive Design. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 321, 7 pages. doi:10.1145/3613905.3650997
- [9] Beni Chugh and Pranjal Jain. 2021. Unpacking Dark Patterns: Understanding Dark Patterns and Their Implications for Consumer Protection in the Digital Economy. *RGNUL Student Research Review Journal* 7 (2021), 23.
- [10] Linda Di Geronimo, Larissa Braz, Enrico Fregnan, Fabio Palomba, and Alberto Bacchelli. 2020. UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, New York, NY, USA. doi:10.1145/3313831.3376600
- [11] Chloe Egtebas, Gudrun Klinker, Susanne Boll, and Marion Koelle. 2023. Co-Speculating on Dark Scenarios and Unintended Consequences of a Ubiquitous (ly) Augmented Reality. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. ACM, Pittsburgh PA USA, 2392–2407. doi:10.1145/3563657.3596073
- [12] European Data Protection Board. 2022. *Guidelines 3/2022 on Dark patterns in Social Media Platform Interfaces: How to Recognise and Avoid Them*. European Data Protection Board, Brussels, Belgium. https://edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-032022-deceptive-design-patterns-social-media_en
- [13] Dan Fitton and Janet C. Read. 2019. Creating a Framework to Support the Critical Consideration of Dark Design Aspects in Free-to-Play Apps. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*. ACM, Boise ID USA, 407–418. doi:10.1145/3311927.3323136
- [14] Alexandre Fleury. 2011. Drawing as a user experience research tool. In *Proceedings of the 11th Danish Human-Computer Interaction Research Symposium (DHRS2011)*. Department of IT Management, Copenhagen Business School (CBS), Frederiksberg, Denmark, 14–17. <http://openarchive.cbs.dk/handle/10398/8359>
- [15] Colin M. Gray, Jingle Chen, Shruthi Sai Chivukula, and Liyang Qu. 2021. End User Accounts of Dark Patterns as Felt Manipulation. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), Article 372. doi:10.1145/3479516
- [16] Colin M. Gray and Shruthi Sai Chivukula. 2019. Ethical Mediation in UX Practice. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19, Paper 178). Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3290605.3300408
- [17] Colin M. Gray, Shruthi Sai Chivukula, Kerstin Bongard-Blanchy, Arunesh Mathur, Johanna Gunawan, and Brennan Schaffner. 2023. Emerging Transdisciplinary Perspectives to Confront Dark Patterns. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. ACM, New York, NY, USA, 4 pages. doi:10.1145/3544549.3583745
- [18] Colin M. Gray, Shruthi Sai Chivukula, and Ahreum Lee. 2020. What Kind of Work Do "Asshole Designers" Create? Describing Properties of Ethical Concern on Reddit. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven, Netherlands) (DIS '20). Association for Computing Machinery, New York, NY, USA, 61–73. doi:10.1145/3357236.3395486
- [19] Colin M. Gray, Johanna T. Gunawan, René Schäfer, Nataliia Bielova, Lorena Sanchez Chamorro, Katie Seaborn, Thomas Mildner, and Hauke Sandhaus. 2024. Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 482, 6 pages. doi:10.1145/3613905.3636310
- [20] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). dl.acm.org, New York, NY, USA, 534:1–534:14. doi:10.1145/3173574.3174108
- [21] Colin M. Gray, Craig M. Macdonald, Carine Lallemand, Alannah Oleson, Anna R. L. Carter, Olivier St-Cyr, and Caroline Pitt. 2023. EduCHI 2023: 5th Annual Symposium on HCI Education. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23,

- Article 347). Association for Computing Machinery, New York, NY, USA, 1–5. doi:10.1145/3544549.3573790
- [22] Colin M. Gray, Cristiana Santos, and Nataliia Bielova. 2023. Towards a Preliminary Ontology of Dark Patterns Knowledge. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 286, 9 pages. doi:10.1145/3544549.3585676
- [23] Colin M. Gray, Cristiana Santos, Nataliia Bielova, Michael Toth, and Damian Clifford. 2021. Dark Patterns and the Legal Requirements of Consent Banners: An Interaction Criticism Perspective. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (CHI'21). ACM Press, New York, NY, USA, Article 172, 18 pages. doi:10.1145/3411764.3445779
- [24] Colin M. Gray, Cristiana Santos, Nicole Tong, Thomas Mildner, Arianna Rossi, Johanna Gunawan, and Caroline Sindors. 2023. Dark Patterns and the Emerging Threats of Deceptive Design Practices. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI EA '23). Association for Computing Machinery, New York, NY, USA, 4 pages. doi:10.1145/3544549.3583173
- [25] Colin M. Gray, Cristiana Santos, Nataliia Bielova, and Thomas Mildner. 2024. An Ontology of Dark Patterns Knowledge: Foundations, Definitions, and a Pathway for Shared Knowledge-Building. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 289, 22 pages. doi:10.1145/3613904.3642436
- [26] Johanna Gunawan, David Choffnes, Woodrow Hartzog, and Christo Wilson. 2021. A Comparative Study of Dark Patterns Across Mobile and Web Modalities. In *Proceedings of the ACM on Human-Computer Interaction* (Virtual Event, Canada) (CSCW '21, Vol. 5). Association for Computing Machinery, New York, NY, USA, Article 377, 29 pages. Issue CSCW2. doi:10.1145/3479521
- [27] Johanna Gunawan, David Choffnes, Woodrow Hartzog, and Christo Wilson. 2021. The COVID-19 Pandemic and the Technology Trust Gap. *Seton Hall Law Review* 51, 3 (Sept. 2021), 29 pages. Issue 5. <https://scholarship.shu.edu/shlr/vol51/iss5/3>
- [28] Johanna Gunawan, Cristiana Santos, and Irene Kamara. 2022. Redress for Dark Patterns Privacy Harms? A Case Study on Consent Interactions. In *Proceedings of the 2022 Symposium on Computer Science and Law* (CSLAW '22). Association for Computing Machinery, New York, NY, USA, 181–194. doi:10.1145/3511265.3550448
- [29] Brignull Harry. 2023. *Deceptive Patterns: Exposing the Tricks Tech Companies Use to Control You*. Testimonium Ltd, London, UK.
- [30] Shun Hidaka, Sota Kobuki, Mizuki Watanabe, and Katie Seaborn. 2023. Linguistic Dead-Ends and Alphabet Soup: Finding Dark Patterns in Japanese Apps. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 3, 13 pages. doi:10.1145/3544548.3580942
- [31] Mitsuki Hosoya, Hiroaki Yamaura, Satoshi Nakamura, Makoto Nakamura, Eiji Takamatsu, and Yujiro Kitaide. 2019. Does the Pop-Out Make an Effect in the Product Selection of Signage Vending Machine?. In *Human-Computer Interaction – INTERACT 2019* (Paphos, Cyprus). Springer-Verlag, Berlin, Heidelberg, 24–32. doi:10.1007/978-3-030-29384-0_2
- [32] Woo Gon Kim, Souji Gopalakrishna Pillai, Kavitha Haldorai, and Wasim Ahmad. 2021. Dark patterns used by online travel agency websites. *Annals of Tourism Research* 88 (2021), 103055. doi:10.1016/j.annals.2020.103055
- [33] Yuichiro Kinoshita, Yuto Sekiguchi, Riho Ueki, Kouta Yokoyama, and Satoshi Nakamura. 2023. Do People Tend to Select a Delayed Item?. In *Human-Computer Interaction* (Copenhagen, Denmark). Springer-Verlag, Berlin, Heidelberg, 397–407. doi:10.1007/978-3-031-35599-8_26
- [34] Monica Kowalczyk*, Johanna Gunawan*, Daniel Dubois, David Choffnes, Woodrow Hartzog, and Christo Wilson. 2023. Understanding Dark Patterns in Home IoT Devices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, 27 pages. doi:10.1145/3544548.3581432
- [35] Alexander Löbel, René Schäfer, Hanna Püschel, Esra Güney, and Ulrike Meyer. 2024. Access Your Data... if You Can: An Analysis of Dark Patterns Against the Right of Access on Popular Websites. In *Privacy Technologies and Policy: 12th Annual Privacy Forum, APF 2024, Karlstad, Sweden, September 4–5, 2024, Proceedings* (Karlstad, Sweden). Springer-Verlag, Berlin, Heidelberg, 23–47. doi:10.1007/978-3-031-68024-3_2
- [36] Kai Lukoff, Alexis Hiniker, Colin M. Gray, Arunesh Mathur, and Shruthi Sai Chivukula. 2021. What can CHI do about dark patterns?. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama Japan). ACM, New York, NY, USA, 6 pages. doi:10.1145/3411763.3441360
- [37] Francisco Lupiáñez-Villanueva, Alba Boluda, Francesco Bogliacino, Giovanni Liva, Lucie Lechardey, and Teresa Rodríguez de las Heras Ballell. 2022. *Behavioural study on unfair commercial practices in the digital environment : dark patterns and manipulative personalisation : final report*. Publications Office of the European Union, Brussels, Belgium. doi:10.2838/859030
- [38] Maximilian Maier and Rikard Harr. 2020. Dark design patterns: An end-user perspective. 16, 2 (2020), 170–199. <https://ht.csr-pub.eu/index.php/ht/article/view/6>
- [39] Arunesh Mathur, Gunes Acar, Michael J. Friedman, Elena Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (Nov. 2019), Article No. 81. doi:10.1145/3359183
- [40] Arunesh Mathur, Mihir Kshirsagar, and Jonathan Mayer. 2021. What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3411764.3445610
- [41] Marcela Mattiuzzo and Paula Pedigoni Ponce. 2024. Power through design – dark patterns, personalization and the emergence of TikTok. *International Review of Law, Computers & Technology* (July 2024), 1–25. doi:10.1080/13600869.2024.2351672
- [42] Thomas Mildner, Orla Cooney, Anna-Maria Meck, Marion Bartl, Gian-Luca Savino, Philip R Doyle, Diego Garaialde, Leigh Clark, John Sloan, Nina Wenig, Rainer Malaka, and Jasmin Niess. 2024. Listening to the Voices: Describing Ethical Caveats of Conversational User Interfaces According to Experts and Frequent Users. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, 2024-05-11). ACM, New York, NY, USA, 1–18. doi:10.1145/3613904.3642542
- [43] Thomas Mildner, Philip Doyle, Gian-Luca Savino, and Rainer Malaka. 2022. Rules Of Engagement: Levelling Up To Combat Unethical CUI Design. In *4th Conference on Conversational User Interfaces*. ACM, Glasgow United Kingdom, 1–5. doi:10.1145/3543829.3544528
- [44] Thomas Mildner, Merle Freye, Gian-Luca Savino, Philip R Doyle, Benjamin R Cowan, and Rainer Malaka. 2023. Defending Against the Dark Arts: Recognising Dark Patterns in Social Media. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 2362–2374. doi:10.1145/3563657.3595964
- [45] Thomas Mildner and Gian-Luca Savino. 2021. Ethical User Interfaces: Exploring the Effects of Dark Patterns on Facebook. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–7. doi:10.1145/3411763.3451659
- [46] Thomas Mildner, Gian-Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23, Article 192). Association for Computing Machinery, New York, NY, USA, 1–15. doi:10.1145/3544548.3580695
- [47] Carol Moser, Sarita Y. Schoenebeck, and Paul Resnick. 2019. Impulse Buying: Design Practices and Consumer Needs. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–15. doi:10.1145/3290605.3300472
- [48] OECD. 2022. *Dark commercial patterns*. Number 336 in OECD Digital Economy Papers. OECD Publishing, Paris, France. 96 pages. doi:10.1787/44f5e846-en
- [49] OECD. 2023. *Consumer Vulnerability in the Digital Age*. Number 355 in OECD Digital Economy Papers. OECD Publishing, Paris, France. 80 pages. doi:10.1787/4d013cc5-en
- [50] Ministry of Consumer Affairs. 2023. Central Consumer Protection Authority issues 'Guidelines for Prevention and Regulation of Dark Patterns, 2023' for prevention and regulation of dark patterns listing 13 specified dark patterns. <https://piib.gov.in/PressReleaseSelfFramePage.aspx?PRID=1983994>
- [51] Ministry of Consumer Affairs. 2023. The Guidelines for Prevention and Regulation of Dark Patterns. <https://consumeraffairs.nic.in/theconsumerprotection/guidelines-prevention-and-regulation-dark-patterns-2023>
- [52] Kentrell Owens, Johanna Gunawan, Pardis Emami-Naeini, David Choffnes, Tadayoshi Kohno, and Franziska Roesner. 2022. Exploring Deceptive Design Patterns in Voice Interfaces. In *Proceedings of the 2022 European Symposium on Usable Security* (Karlsruhe, Germany) (EuroUSEC '22). Association for Computing Machinery, New York, NY, USA, 64–78. doi:10.1145/3549015.3554213
- [53] Lorena Sanchez Chamorro, Carine Lallemand, and Colin M. Gray. 2024. "My Mother Told Me These Things are Always Fake" - Understanding Teenagers' Experiences with Manipulative Designs. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 1469–1482. doi:10.1145/3643834.3660704
- [54] René Schäfer, Paul Miles Preuschoff, and Jan Borchers. 2023. Investigating Visual Countermeasures Against Dark Patterns in User Interfaces. In *Proceedings of Mensch Und Computer 2023* (Rapperswil, Switzerland) (MuC '23). Association for Computing Machinery, New York, NY, USA, 161–172. doi:10.1145/3603555.3603563
- [55] René Schäfer, Paul Miles Preuschoff, René Röpke, Sarah Sahabi, and Jan Borchers. 2024. Fighting Malicious Designs: Towards Visual Countermeasures Against Dark Patterns. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 296, 13 pages. doi:10.1145/3613904.3642661

- [56] René Schäfer, Sarah Sahabi, Annabell Brocker, and Jan Borchers. 2024. Growing Up With Dark Patterns: How Children Perceive Malicious User Interface Designs. In *Nordic Conference on Human-Computer Interaction* (Uppsala, Sweden) (*NordiCHI 2024*). Association for Computing Machinery, New York, NY, USA, Article 25, 17 pages. doi:10.1145/3679318.3685358
- [57] Brennan Schaffner, Neha A. Lingareddy, and Marshini Chetty. 2022. Understanding Account Deletion and Relevant Dark Patterns on Social Media. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (Nov. 2022), 1–43. doi:10.1145/3555142
- [58] Katie Seaborn and Weichen Joe Chang. 2024. Another Subtle Pattern: Examining Demographic Biases in Dark Patterns and Deceptive Design Research. In *Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices Workshop at CHI Conference on Human Factors in Computing Systems (DDP CHI '24)*. CEUR-WS.org, online, 10 pages. <https://chi2024.darkpatternsresearchandimpact.com/>
- [59] Katie Seaborn, Tatsuya Itagaki, Mizuki Watanabe, Yijia Wang, Ping Geng, Takao Fujii, Yuto Mandai, Miu Kojima, and Suzuka Yoshida. 2024. Deceptive, Disruptive, No Big Deal: Japanese People React to Simulated Dark Commercial Patterns. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 95, 8 pages. doi:10.1145/3613905.3651099
- [60] Riho Ueki, Kouta Yokoyama, and Satoshi Nakamura. 2023. Does the Type of Font Face Induce the Selection?. In *Human-Computer Interaction: Thematic Area, HCI 2023, Held as Part of the 25th HCI International Conference, HCII 2023, Copenhagen, Denmark, July 23–28, 2023, Proceedings, Part II* (Copenhagen, Denmark). Springer-Verlag, Berlin, Heidelberg, 497–510. doi:10.1007/978-3-031-35599-8_33
- [61] Xian Wang, Lik-Hang Lee, Carlos Bermejo Fernandez, and Pan Hui. 2023. The Dark Side of Augmented Reality: Exploring Manipulative Designs in AR. *International Journal of Human-Computer Interaction* 0, 0 (March 2023), 1–16. doi:10.1080/10447318.2023.2188799
- [62] Kota Yokoyama, Satoshi Nakamura, and Shota Yamanaka. 2021. Do Animation Direction and Position of Progress Bar Affect Selections?. In *Human-Computer Interaction – INTERACT 2021* (Bari, Italy). Springer-Verlag, Berlin, Heidelberg, 395–399. doi:10.1007/978-3-030-85607-6_45
- [63] José P. Zagal, Staffan Björk, and Chris Lewis. 2013. Dark Patterns in the Design of Games. In *Proceedings of the 8th International Conference on the Foundations of Digital Games (FDG 2013)*. Society for the Advancement of the Science of Digital Games, Santa Cruz, CA, USA, 8. <https://urn.kb.se/resolve?urn=urn:nbn:se:ri:diva-24252>